ReDataX: a decision-making system

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1 Introduction

Financial platforms, whether cryptocurrency exchanges, brokers, or payment systems, increasingly have to manage risk in real time. One of the most difficult and costly forms of that risk is adverse selection. It appears when a platform temporarily carries market exposure from client trades before the position is fully hedged or offset. During that interval, price can move against the platform's open position and produce a loss.

In classical market microstructure, the issue arises because part of the order flow may contain information about future price movements. If a platform executes that flow at the current price, the market may move adversely after the trade. The resulting loss is not always visible from the trade itself beforehand [3, 4].

At first glance, the natural answer is to forecast price at short time scales. Price formation is stochastic, however, and even short-horizon prediction is far from easy: noise is large, useful signal is weak, and a good predictive metric does not automatically translate into positive economic value. ReDataX takes a different route. It analyzes the state of market flow, tries to identify the part of that flow that is costly to the platform, infers when risk is emerging, and proposes an intervention only when acting appears worthwhile.

The original intuition comes from statistical physics. A gas in a vessel cannot be described usefully by following every particle microscopically: for one mole of matter, one would need to solve roughly 6×10^{23} equations of motion, and even if that were possible, it would not be obvious what to do with so much information. Instead, physics uses averaged quantities that retain what matters about the state of the system. In ReDataX, that means constructing effective features of market flow at several time scales.

More specifically, the idea is inspired by Kadanoff block transformations [1] and the renormalization-group viewpoint: some fine detail is discarded, while the information needed to describe behavior at a coarser scale is retained. The general renormalization-group framework in statistical physics and stochastic dynamics is discussed, for example, in [2].

The project's central hypothesis is straightforward: if market activity is represented by a feature set Ω that captures the imbalance between aggressive buys and sells across several time scales, then that representation may contain information about a future adverse movement.

The hypothesis is tested on public Binance `aggTrades` data for BTCUSDT and ETHUSDT.

2 Objective

The main objective of ReDataX is to build a protocol for evaluating market flow from the perspective of adverse selection and to test whether that protocol can support intervention decisions.

First, the project asks whether a multiscale description of flow adds information beyond a single fixed scale.

Second, it asks whether relationships between markets help. BTCUSDT and ETHUSDT do not exist in isolation: both have a dollar-like quote currency, respond to similar market events, and are connected through the ETHBTC cross-rate. The experiment tests whether cross-market features improve the description of the current flow state.

Third, if the main hypothesis is correct and a predictive signal really exists, the project asks whether that signal can be turned into a decision. The model should help select the trades for which intervention is economically justified.

3 Business relevance

When processing a trade, a financial platform can broadly do one of two things: hedge the position immediately in the external market and pay a fee, spread, or slippage, or carry the risk for a while. As long as the position remains unprotected, price may move against the platform. If the platform sells an asset to a client and the price then rises, it sold too cheaply relative to the future price. If it buys the asset from a client and the price then falls, it paid too much. Both are examples of adverse selection.

A similar problem appears in market making, where a participant quotes bid and ask prices while simultaneously managing execution risk and inventory [5]. ReDataX does not reproduce that model, but it follows the same general logic: a risk estimate becomes meaningful only when it is connected to an action.

In this study, *intervention* is an umbrella term for actions that may reduce risk. It may mean immediate hedging, changing trade routing, temporarily adjusting the spread, reducing inventory exposure, or using some other control. From this point onward, it is convenient to treat an intervention as an operation that reduces risk at a known cost.

The project does not claim to reproduce the internal algorithm of any financial institution. Actual hedging, routing, and client-flow parameters are private. ReDataX is a research project inspired by some of the operating principles associated with Revolut.

4 System boundaries

The test dataset consists of public Binance aggregate trades for BTCUSDT and ETHUSDT. These data do not, and cannot, contain information available only to a bank, broker, or payment platform: the institution's real client flow, client categories, current inventory, internalization strategy, exact hedging decisions, execution quality, routing logic, client response to price changes, real fees, spreads, or execution costs.

This leads to an important qualification. Monetary values in this study, including protected value and net policy value, are not the actual profit of any organization. They estimate how much value could potentially have been protected under the stated assumptions.

5 What do we need from the model?

Definition 5.1. An event i occurring at time t_i is called an *observation.*

At the level of one observation, the output of the system is a decision made at time t_i. For every observation, a feature vector

$$x_i = x(t_i)$$

is constructed using only information available at or before the decision time. It may include trade count, average trade size, aggressive buy-sell imbalance, features at different time scales, synchronous features from a neighboring market, and other quantities. The essential requirement is simple: no feature may use future information.

Version 1.0 evaluates decisions at a fixed 10-second stride. The model therefore does not have to decide on every individual trade; it evaluates the flow state at discrete times. A future ReDataX version may make this stride dynamic.

6 Core definitions

This section introduces the main quantities used in the prediction problem and the economic evaluation of a policy. Version 1.0 fixes the base-scenario parameters at

$$\rho_{\text{int}} = 0.25, \qquad \rho_{\text{mit}} = 0.50, \qquad c_{\text{act}} = 0.50 \text{ bps}, \qquad m_{\text{BE}} = 4.00 \text{ bps}, \qquad B = 10\%.$$

Definition 6.1. The *forecast horizon* H is the length of the interval over which future risk is evaluated.

If a decision is made at t_i, for example, its outcome is evaluated at $t_i + H$.

Definition 6.2. The *markout* of observation i at horizon H, denoted $M_i(H)$, is the price change between decision time t_i and future time $t_i + H$, measured in basis points.

Formally, it is the difference between the future and current prices, with orientation chosen so that positive markout corresponds to a loss for the platform and negative markout to a gain. Under this convention, risk is a monotonically increasing function of $M_i(H)$. The exact aggressor-aligned markout implementation is given in `docs/ml/targets.md`.

Definition 6.3. An *adverse-selection event* at horizon H is

$$Y_i(H) = \mathbf{1}\{M_i(H) > 0\}, \tag{1}$$

where $\mathbf{1}(\cdot)$ is the indicator function.

If future markout is positive, the observation is treated as adverse; otherwise, the event did not occur. This is the logic used by the M0–M3 classification models, which learn to predict the probability of an adverse movement.

Not all movements matter equally. Every intervention has a cost, and if the movement is too small, that cost may exceed the value protected. A threshold handles this distinction:

$$Y_i^{\text{BE}}(H) = \mathbf{1}\{M_i(H) > m_{\text{BE}}\}, \tag{2}$$

where m_{BE} is break-even markout. It is constant in the present study, although a future version could make it dynamic.

For the binary M0–M3 models, a 1-basis-point move and a 100-basis-point move are indistinguishable, even though their economic consequences are very different.

Definition 6.4. The *loss magnitude* of adverse markout at forecast horizon H is

$$S_i(H) = \max(M_i(H), 0). \tag{3}$$

Introducing magnitude lets us split the task into two parts: estimating the probability of an adverse event and estimating loss size conditional on that event:

$$\mathbb{P}(Y_i = 1 \mid x_i) \qquad \mathbb{E}[S_i \mid Y_i = 1, x_i]. \tag{4}$$

By the definition of conditional expectation,

$$\mathbb{E}[S_i \mid x_i] = \mathbb{P}(Y_i = 1 \mid x_i) \times \mathbb{E}[S_i \mid Y_i = 1, x_i]. \tag{5}$$

This is the logic behind M4 and M5. M4 estimates $\mathbb{E}[S_i \mid x_i]$ directly, while M5 estimates event probability and conditional loss magnitude separately.

Definition 6.5. A *toxic flow*[1] is a flow after which price is likely to move against the platform. Observation i is called *toxic* when its future markout is positive: $M_i(H) > 0$.

For example, suppose a client wants to place a large trade, but executing it as one order would reveal the full demand at a fixed price. The client can split it into many smaller orders to conceal the intention, potentially moving the market just after the market maker has filled those orders at a favorable price. For the toxicity label, what matters is the direction of the subsequent move, not its size.

Definition 6.6. The *internalization rate* ρ_{int} is the share of notional that the platform hypothetically retains on its own balance sheet:

$$\rho_{\text{int}} \in [0, 1]. \tag{6}$$

The current ReDataX version uses $\rho_{\text{int}} = 0.25$.

Definition 6.7. The *mitigation efficiency* ρ_{mit} is the share of adverse exposure prevented by an intervention:

$$\rho_{\text{mit}} \in [0, 1]. \tag{7}$$

The current ReDataX version uses $\rho_{\text{mit}} = 0.50$.

Definition 6.8. The *action cost* c_{act} is the intervention cost in basis points of notional touched.
The current ReDataX version uses $c_{\text{act}} = 0.50$ bps.

Every intervention has a price: exchange fees, spreads, operating costs, or market impact. If $c_{\text{act}} = 0.50$ bps, for example, acting on a \$100,000 trade costs \$5. The corresponding break-even markout is

$$m_{\text{BE}} = \frac{c_{\text{act}}}{\rho_{\text{mit}}\rho_{\text{int}}} = \frac{0.50}{0.50 \times 0.25} = 4.00 \text{ bps}.$$

Definition 6.9. *Notional touched* is the share of total notional on which the policy intervenes:

$$f_{\text{act}} = \frac{\sum_i a_i N_i}{\sum_i N_i},$$

where $a_i \in \{0, 1\}$ is the policy decision and N_i is the notional of observation i.
The current ReDataX version imposes $f_{\text{act}} \leq B$, with $B = 10\%$.

For every observation, the intervention policy chooses

$$a_i = \begin{cases} 1, & \text{intervene on this trade, for example by hedging it,} \\ 0, & \text{do not intervene.} \end{cases}$$

Intervening on every trade would be both expensive and risky. This is why the budget constraint exists: the policy may cover only a specified share of total notional. That share is the notional touched.

The constraint $f_{\text{act}} \leq B$ means that no more than 10% of total notional may be subject to action. In other words, it limits the share of turnover on which intervention is permitted.

[1] The term "toxic" is not universal. In this study, it labels observations with positive future markout. The market-microstructure literature uses flow toxicity in a closely related sense when discussing adverse selection faced by liquidity providers.

Definition 6.10. Let L_i denote the adverse exposure that arises if no intervention takes place:

$$L_i = N_i \rho_{\text{int}} \frac{S_i}{10^4}.$$

The *exposure capture ratio* is the share of potential adverse exposure contained in the observations selected by the policy:

$$C = \frac{\sum_i a_i L_i}{\sum_i L_i}, \tag{8}$$

where $a_i \in \{0, 1\}$ is the policy decision.

If $C = 30\%$, the selected interventions cover 30% of all adverse exposure. The share actually prevented also depends on mitigation efficiency ρ_{mit}.

Definition 6.11 (Risk concentration)**.** *Risk concentration* is the ratio of captured exposure to notional touched:

$$R = \frac{C}{f_{\text{act}}}.$$

If $R > 1$, the policy captures a larger share of adverse exposure than its share of notional touched. $R = 1$ corresponds to a selection that, on average, captures exposure in proportion to notional touched. If $R < 1$, random selection would be more efficient.

7 Prediction problem

We now have the basic vocabulary: markout, toxicity, exposure, and the budget constraint. The next step is to define the models that estimate risk from the current market state.

7.1 Binary classification

The first model family, M0–M3, solves a binary-classification problem. These models estimate the probability of a future adverse event:

$$p_i(H) \equiv \mathbb{P}\left(Y_i(H) = 1 \mid x_i\right).$$

Here $Y_i = 1$ means that markout is positive, and x_i is the feature vector describing the market state at time t_i.

The model outputs p_i, and observations are ranked in descending order of that probability. Intervention is assigned to the highest-ranked observations until the notional-touched budget is exhausted.

The primary metric is Average Precision (AP). It measures how effectively the model lifts positive-label observations toward the top of the ranking. Higher AP raises the chance of capturing more adverse exposure under a fixed budget, although the economic outcome also depends on notional, markout magnitude, and intervention cost.

7.2 Loss-magnitude estimation

If the classification models produce a useful ranking, the next step is to estimate the magnitude of potential loss. M4 therefore estimates future positive adverse markout:

$$\widehat{S}_i^{\text{direct}}(H) \approx \mathbb{E}\left[S_i(H) \mid x_i\right].$$

This regression model predicts S_i as a continuous quantity in basis points. If it returns $\widehat{S}_i = 12.5$ bps, the interpretation is that the expected loss associated with the observation is 12.5 basis points. The simplicity comes at a cost: M4 does not distinguish a small probability of a large loss from a large probability of a small loss. Direct regression may assign the same value to both, even though the two cases can call for different decisions.

7.3 Hurdle formulation

M5 decomposes the problem differently. It estimates the probability that a loss occurs at all,

$$p_i(H) = \Pr\left(S_i(H) > 0 \mid x_i\right),$$

and the expected loss conditional on that event:

$$\mu_i(H) = \mathbb{E}\left[S_i(H) \mid S_i(H) > 0, x_i\right].$$

The expected markout score is the product of the two components:

$$\widehat{S}_i^{\text{hurdle}}(H) = p_i(H) \cdot \mu_i(H).$$

Compared with M4, M5 introduces a two-part view of the decision: event probability and loss magnitude. Keeping the two pieces visible makes the policy more flexible and, potentially, more robust.

8 Decision problem

A predictive score is not an intervention policy by itself. Models provide probabilities or magnitude estimates; they do not act. Their outputs must therefore be converted into action decisions:

$$a_i = \pi(x_i, \widehat{p}_i, \widehat{\mu}_i; \theta),$$

where π is the policy, θ contains its thresholds and budget parameters, and $a_i = 1$ means that the observation has been selected for intervention.

The policy must satisfy the notional constraint

$$\frac{\sum_i a_i N_i}{\sum_i N_i} \leq B.$$

9 Modeling intervention economics

The business is interested in more than model scores and decision rules. Markouts and probabilities therefore have to be translated into concrete economic quantities.

9.1 Loss prevented

For every observation i on which the policy acts ($a_i = 1$), prevented loss is

$$G_i = a_i N_i \rho_{\text{int}} \rho_{\text{mit}} \frac{S_i(H)}{10^4}.$$

Here $a_i \in \{0, 1\}$ is the intervention decision, N_i is trade notional, $\rho_{\text{int}} = 25\%$ is the hypothetical internalization share, $\rho_{\text{mit}} = 50\%$ is the share of loss prevented by intervention, and S_i is loss magnitude.

Suppose we intervene on a \$100,000 trade, the adverse move is 50 bps, and internalization and mitigation efficiency are 25% and 50%, respectively. Then

$$G_i = 1 \times 100{,}000 \times 0.25 \times 0.50 \times \frac{50}{10{,}000} = \$62.50.$$

Without intervention, adverse exposure would have been \$125. With 50% mitigation efficiency, the intervention protects \$62.50. Total prevented loss over a period is

$$G = \sum_i G_i.$$

9.2 Intervention cost

Intervention cost is modeled as a fixed fraction of trade notional:

$$K_i = a_i N_i \frac{c_{\text{act}}}{10^4}.$$

For a \$100,000 trade, for example,

$$K_i = 100{,}000 \times \frac{0.50}{10{,}000} = \$5.00.$$

The total cost of all interventions is

$$K = \sum_i K_i.$$

9.3 Net protected value

Net protected value is protected value minus intervention cost:

$$V = G - K.$$

A value $V < 0$ means that protected exposure does not cover the cost of acting. In the example from Sections 9.1 and 9.2,

$$V = \$62.50 - \$5.00 = \$57.50.$$

To compare results across markets and samples, net protected value is normalized per \$1 million of turnover:

$$V_{\$1M} = \frac{V}{\sum_i N_i} \times 10^6.$$

This quantity reports policy value for every million dollars of turnover.

9.4 Break-even action cost

It is useful to know how robust a policy is to higher costs. The break-even action cost is the intervention cost at which net value becomes zero:

$$c_{\text{BE}} = \frac{G}{\sum_i a_i N_i} \times 10^4.$$

If the current cost c_{act} is below c_{BE}, the policy remains positive under the scenario. Its cost headroom is

$$h_c = c_{\text{BE}} - c_{\text{act}}.$$

The larger h_c, the more room the policy has for cost increases.

9.5 Benefit-cost ratio

The benefit-cost ratio is

$$\mathrm{BCR} = \frac{G}{K}, \quad K > 0.$$

It tells us how many dollars of gross protected value are obtained for each dollar spent on intervention.

A value $\mathrm{BCR} > 1$ means that the policy creates positive economic value under the selected scenario assumptions.

10 Policy set

The study evaluates five policy types.

10.1 P0 - No action

P0 never intervenes:

$$a_i = 0 \quad \forall i.$$

It is the zero baseline against which the other policies are compared.

10.2 P1 - Probability-budget policy

P1 uses the M0–M3 classification models. For every observation, it calculates the probability of an adverse event p_i. Observations are ranked by decreasing p_i, and the policy acts on the highest-ranked ones until budget B is exhausted. In other words, it spends the permitted notional-touched share on observations with the largest predicted probability of an adverse movement.

10.3 P2 - Direct economic policy

Instead of a probability, P2 uses the expected-loss estimate from M4, $\widehat{S}_i^{\text{direct}}$. Observations are ranked by decreasing $\widehat{S}_i^{\text{direct}}$, and the policy considers the top $B\%$. In practice, the score is then evaluated against action cost and the notional-touched budget.

10.4 P3 - Hurdle economic policy

P3 uses M5, which estimates the probability and possible magnitude of an adverse event separately. Those quantities are used to calculate the expected economic effect of intervention. The policy then selects observations that clear its economic threshold and fit within the notional-touched budget.

10.5 P4 - Oracle upper bound

P4 is an idealized policy that uses realized future information. Rather than relying on model predictions, it ranks observations by the true loss S_i, which is known only after the fact.

P4 cannot be deployed in a real system, but it provides an upper bound on the value attainable if the future were known. Comparing P3 with P4 shows how much potential value remains unprotected.

Detailed policy definitions are provided in `docs/decision_policies/`.

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